

AP CSP | Lesson 03: Variables & Sequence

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: B

B is correct: $16 + 8 + 1 = 25$. A would be 00010011, using the 16, 2 and 1 weights. C would be 00011010, using 16, 8 and 2. D would be 00110001, using 32, 16 and 1. Leading zeros do not change the integer value.

Q2. Answer: C

C is correct: the count enters the program and the total is displayed. A mistakes a value already fixed inside the code for the stated user input, and treats the received count as output. B reverses the actual input and output. D confuses purpose with data and an internal processing operation with output.

Q3. Answer: A

A is correct: y receives a copy of 8. The third line changes x to 13 but does not assign to y. B incorrectly treats y as a live link to x. C gives the amount added to x, not the value of y. D adds the old and new values of x even though no statement performs that addition or stores it in y.

Q4. Answer: D

D is correct: multiplication occurs first, so $6 + 4 \times 3 = 6 + 12 = 18$. A evaluates $(6 + 4) \times 3$, introducing parentheses that the code does not contain. B treats the multiplication as addition: $6 + 4 + 3$. C correctly multiplies 4 by 3 but omits the addition of 6.

Q5. Answer: B and C

B is correct because the last line evaluates $10 - 3$ and stores 7 in points. C is correct because saved received 10 before that change. A mistakes the assignment arrow for an equality comparison. D mistakes copying a current numeric value for creating an ongoing link. Award one mark only for B and C together with no other choices.

Q6. Answer: B

B is correct: total is assigned $3 \times 12 = 36$ before price becomes 10. The later assignment to price does not recalculate total. A uses the final price instead of the price when total was calculated. C reports the final value of price, but DISPLAY uses total. D reports the initial price and omits the multiplication.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: (a) After line 2: (3, 8). After line 3: (16, 8). (b) 16, then 7. (c) Line 4 assigns only to b. The earlier calculation copied the value 16 into a; changing b later does not recalculate or replace a.

1 mark: Gives (a, b) = (3, 8) after line 2.

1 mark: Gives (a, b) = (16, 8) after line 3.

1 mark: Gives both displayed values in the correct order: 16, then 7.

1 mark: Explains that line 4 assigns to b only and does not recalculate the value already stored in a.

Q8. 4 marks

Answer: (a) 51. (b) 60. (c) $4 * 12 + 3$ matches a fee once per order. The other expression charges the fee for each of the four tickets. (d) $total \leftarrow tickets * price + fee$.

1 mark: Evaluates $4 * 12 + 3$ as 51.

1 mark: Evaluates $4 * (12 + 3)$ as 60.

1 mark: Selects $4 * 12 + 3$ and explains once per order versus once per ticket.

1 mark: Writes $total \leftarrow tickets * price + fee$, or an equivalent correctly parenthesized assignment using the supplied variables.

Q9. 4 marks

Answer: 1. Receive the number of sessions. 2. Multiply the session count by 8 to obtain the session subtotal. 3. Add 5 once to that subtotal to obtain the total. 4. Display the total.

1 mark: Explicitly receives or reads the number of sessions before using it.

1 mark: Multiplies the session count by 8, after receiving the count.

1 mark: Adds a single fee of 5 to the session subtotal, or clearly describes $count * 8 + 5$.

1 mark: Displays the calculated total after calculating it. A combined correct calculation is acceptable.

Q10. 4 marks

Answer: (a) Actual output: 24. Required output: 20. (b) $price \leftarrow 12$

$price \leftarrow 10$

$total \leftarrow 2 * price$

`DISPLAY(total)`

(c) The calculation of total now uses $price = 10$. In the original order it used 12, and changing price afterward did not recalculate total.

1 mark: States the actual output 24.

1 mark: States the required output 20.

1 mark: Writes the original four statements in the order $price \leftarrow 12$; $price \leftarrow 10$; $total \leftarrow 2 * price$; `DISPLAY(total)`, moving one existing line.

1 mark: Explains that total now uses the updated price at calculation time and that a later price change does not automatically recalculate total.

Q11. 4 marks

Answer: (a) first = 14 and second = 6. (b) $(14 + 6) / 2 = 10$. (c) 17, because division is evaluated before addition: $14 + 6 / 2 = 14 + 3 = 17$.

1 mark: Assigns the supplied values in order: first = 14 and second = 6.

1 mark: Gives the original output 10.

1 mark: Gives the modified output 17.

1 mark: Explains that $second / 2$ is evaluated before adding first, or shows $14 + 3 = 17$.

Q12. 4 marks

Answer: (a) $hours \leftarrow INPUT()$

$rate \leftarrow 6$

$fee \leftarrow 4$

$total \leftarrow hours * rate + fee$

`DISPLAY(total)`

(b) Input 1 gives 10. Input 5 gives 34.

1 mark: Receives the hour count with `INPUT()` and stores it in a meaningful variable before use.

1 mark: Assigns 6 and 4 to meaningful named rate and fee variables before using them.

1 mark: Calculates $hours * rate + fee$ and displays the calculated result, using valid assignment and sequence; equivalent extra subtotal steps are acceptable.

1 mark: Gives both expected outputs correctly: 10 for input 1 and 34 for input 5.